

Amir Khajavirad

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PROFESSIONAL SUMMARY

Biomedical engineer with hands-on experience across robotics and embedded systems development, biomedical device testing, and clinical data analysis. Builds and maintains Raspberry Pi based robotic platforms integrating sensors, motors, GUI controls, and networked multi-device communication for laboratory research, and applies statistics and machine learning to clinical and experimental datasets. Comfortable working across hardware, software, and data layers to deliver reliable, reproducible research tools.

TECHNICAL SKILLS

Programming and Systems: Python, SQL, Bash, Linux, Object-Oriented Programming, Multithreading, File-Based IPC, JSON/CSV Data Handling

Embedded and Robotics: Raspberry Pi 4/5, GPIO, SPI, UART/Serial, RFID, HX711 Load Cells, Stepper Motor Control, Sensor Integration, Hardware-Software Integration

Control, GUI and Networking: Sensor-Driven Control Logic, State Management, Safety Interlocks, Python Tkinter Touchscreen GUIs, TCP Socket Communication (Pi-to-Pi Protocol), NetworkManager/nmcli, WPA/WPA2 Enterprise Wi-Fi

Software Reliability and Deployment: Git, GitHub, CI/CD (in progress), Versioned Release Packaging, Preflight Validation, Staging, Backup and Rollback, Runtime Logging

Data Science, ML and Statistics: Python, Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, SHAP, Feature Engineering, PCA/Kernel PCA/LDA, Descriptive Statistics, Confidence Intervals, Risk Ratio, Fisher's Exact Test

Clinical and Biomedical Data: Longitudinal Data Analysis, Repeated Measures, Hematologic Data Analysis, Data Quality Assessment, Baseline Adjustment, Risk Estimation

Engineering and Design: SolidWorks, CAD Modeling, Mechanical Design, Rapid Prototyping

EXPERIENCE

Robotics Engineer (Intern to Part-Time)

Nov 2025 - Present

University of Alberta — Edmonton, AB

- Developed and now lead ongoing development of a fleet of Raspberry Pi based robotic rehabilitation platforms, scaling deployment from 0 to 5 units, with 12 planned within two months.
- Built a Python/Tkinter touchscreen control interface covering robot startup/stop, RFID management, experiment configuration, live monitoring, calibration workflows, and software updates, taking the platform from no GUI to a full operator interface.
- Supervise two co-op mechatronics students within a 10-member lab team, reviewing code and guiding hardware debugging under two direct supervisors.
- Designed and advanced a TCP-based multi-robot (Pi-to-Pi) communication protocol, including RFID transfer, outcome messaging, acknowledgements, and duplicate protection.
- Debugged hardware-software integration issues across GPIO, SPI, UART/serial, RFID, load cells, stepper motors, and Raspberry Pi 4/5 hardware, and refined calibration workflows with operator-guided safety checks.
- Engineered a guarded software deployment workflow with versioned packages, preflight validation, staging, protected backups, and rollback support, and began implementing CI/CD to improve release reliability.
- Designed an on-device Wi-Fi management system (NetworkManager/nmcli) supporting touchscreen network discovery and WPA/WPA2 Enterprise connectivity.

PROJECTS

Clinical Data Analysis — Breast Cancer Supportive Oncology Dataset

Apr 2026 - Jun 2026

- Analyzed a longitudinal dataset of repeated hematologic measurements and chemotherapy-related adverse events across 34 breast cancer patients receiving melatonin or placebo.
- Performed Excel data validation, cleaning, unit normalization, and consistency checks, and corrected inconsistent white blood cell count scaling with an auditable workflow.
- Calculated descriptive statistics, risk estimates, 95% confidence intervals, and Fisher's exact tests for outcomes including neutropenia incidence and treatment-related hospitalization, with reproducible, publication-ready visualizations.

Standardized Biomechanical Testing Platform for Distal Biceps Fixation

Sep 2025 - Dec 2025

Master's capstone project — University of Alberta

- Designed and built a biomechanical testing platform using 3D-printed anatomical bone models and custom fixtures integrated with a BOSE ElectroForce mechanical testing system.
- Implemented cyclic loading and pull-to-failure protocols to evaluate unicortical and bicortical distal biceps fixation, enabling quantitative comparison between methods.
- Presented findings at the University of Alberta engineering research symposium (poster: "Testing of Repair Techniques for Distal Biceps Tendon Tears").

PredictIQ — Football Match Outcome Prediction (XGBoost)

Sep 2025 - Dec 2025

Team project (2 members) — github.com/shervinafzali/PredictIQ

- Co-developed an end-to-end ML pipeline with time-aware validation to prevent data leakage, optimizing an XGBoost model to a macro F1 score of 0.357.
- Applied SHAP to interpret model behavior and communicate feature importance.

Breast Cancer Subtype Classification from Microarray Data

Jan 2025 – Apr 2025

Graduate course project

- Built classification models on a high-dimensional microarray dataset (151 biopsies, 54,676 genes, 6 subtypes), applying PCA, Kernel PCA, and LDA to address multicollinearity.
- Evaluated SVM, Random Forest, KNN, Logistic Regression, and Gaussian Naive Bayes, achieving up to 96% accuracy and 94% macro F1 across 5-run cross-validation.

EDUCATION

Master of Science, Biomedical Engineering

Jan 2024 – Dec 2025

University of Alberta — Edmonton, AB

- Capstone: "Development of a Standardized Biomechanical Testing Setup for Unicortical and Bicortical Distal Biceps Cortical Button Fixation."
- Relevant coursework: Machine Learning in Biomedical Applications, Biomedical Device Design and Technology Development, Applied Machine Learning and AI, Fundamentals of Biomechanical Modelling.

Bachelor of Science, Biomedical Engineering

Sep 2017 – Jun 2022

Islamic Azad University — Tehran, Iran

- Capstone: designed and built an embedded biomedical device (adaptive circadian lighting system) for neonatal incubators.

CERTIFICATIONS

- Linux and SQL — Google, Issued Mar 2026 (Credential ID 2MZC731Z3D7V)
- Google Data Analytics Specialization — Google, Issued Apr 2025 (Credential ID JHLJ6YOKJQ3H)
- Master SolidWorks: Beginner to Advanced — Udemy, Issued Mar 2026

ADDITIONAL EXPERIENCE AND LEADERSHIP

- Recreation Facility Operator, University of Alberta (Sep 2024 – Present) — manages operations in a high-traffic environment, ensuring safety and real-time issue resolution.
- DRA Research Supervisor, University of Alberta — supervises undergraduate research activities and guides system-level problem solving.
- Event volunteer/coordinator, Engineering Graduate Research Symposium and BioAlberta Health and Life Sciences Showcase 2025, University of Alberta.